



SPIN-OFF A · BETA

OrbitDB Relay

Backup, pinning and recovery for local-first data

| The data is yours — we make sure it does not disappear.

[Docs](#)[Source](#)[npm](#)

The problem

Local-first means the data lives on the device. Which is exactly where it gets lost.

- **No server, no backup**

Clear the browser storage, lose the phone, break the laptop — and the database is gone. Without a backend, nobody else is holding a copy.

- **All peers offline = nobody reachable**

A shared list, a project, a chat: while no peer is online, the database is simply not retrievable for anyone joining.

- **Identity is bound to the device**

Passkeys live in one device's secure element. Lose the device and you lose the identity — and with it write access to your own data.

- **Permissions vanish with it**

UCAN delegations and access-control lists are data themselves. Lose them and collaboration breaks, even if the content still exists.

The solution

One node you can rent, self-host, or spin up for a meeting and throw away afterwards. It does not replace a backend — it endures what browsers cannot.

- **Connect** TODAY
libp2p relay with WebSocket, WebRTC-Direct and QUIC, automatic TLS and bootstrap discovery — browsers find each other through NATs.
- **Keep available** TODAY
Replicates OrbitDB oplogs and pins the corresponding IPFS blocks including media while no peer is online.
- **Archive** IN PROGRESS
Dehydration: the database is pushed to Filecoin or Aleph IPFS through the **OrbitDB** ↔ **Storacha Bridge** — hash- and identity-preserving, so it is verifiably the same data.
- **Restore** PLANNED
Hydration: a new device proves its right via a UCAN delegation — and gets back the oplog, the blocks, the database addresses and the permissions.

What gets protected

Not just files. A collaboration environment is four things — miss one and the restore is worthless.

Data

- OrbitDB oplog (every entry, signed)
- IPFS blocks and manifests
- Media: images, attachments

Identity

- Passkey *public keys* and DID documents
- Via **p2pass** and **UCAN Store**
- Private keys never leave the authenticator

Permissions

- UCAN delegations
- Access controllers / per-DID ACLs
- Who may write, who may only read

Environment

- The group's database addresses
- Peer and topic hints
- Bootstrap addresses to rejoin

The relay sees public keys, never private ones. Recovery never means "somebody else can now write as you".

Who it is for

Wherever data matters too much to park it at a provider — and too much to lose.

Teams without IT

- Clubs, practices, law firms, small businesses
- Want neither a SaaS subscription nor a server
- Still need a backup somebody is accountable for

NGOs and newsrooms

- Work where providers get shut down
- A Filecoin archive outlives devices and companies
- Access via delegations instead of accounts

App vendors

- Build local-first and trip over the backup question
- Want to buy pinning and recovery instead of building it
- Embeddable as a service or as a library

Business model

The core stays open source. People pay for operation, storage and accountability — not for access to the code.

- **Managed relay**

One click through the Relay Button, billed by runtime — for a meeting, a project or permanently.

- **Archive quotas**

Storage packages on Filecoin/Aleph with guaranteed restore; storage cost is passed through, the margin sits in operations.

- **Recovery service**

The actual moment of sale: "device gone, data back." Free as self-service, paid when we walk you through it.

- **White label / SLA**

App vendors and companies buy pinning, retention windows and support under their own brand.

Status & next steps

The relay runs under our own demos today. What is missing is the path from "keeps it available" to "brings it back".

- **Beta on npm, docs online** TODAY

`npm i -g orbitdb-relay` — replication, pinning, HTTP API, access controllers and WebAuthn identities included.

- **Archive pipeline** IN PROGRESS

Wire in the Storacha bridge: scheduled dehydration, checksums, restore tests in CI.

- **UCAN-based hydration** PLANNED

Restore against a delegation instead of an account — together with p2pass and UCAN Store.

- **Encrypted archives** PLANNED

Payload encryption, so even the operator of the relay cannot read the content.

ORBITDB RELAY

Let's talk

We are looking for pilot customers with real data and real risk of loss — and partners building local-first apps who need to answer the backup question.

Le Space UG (haftungsbeschränkt) · Nico Krause

contact@le-space.de · local-first.le-space.de

+49 / 87 21 / 5 06 49 96

Docs

Source

npm